

Get A Hold on the Scheduler with CPU Affinity, in Linux

Understanding CPU affinity helps in designing a better application for SMP systems, by binding a certain process to a specific CPU. This article discusses the CPU affinity interface to Linux.

 In a multi-core/multi-processor system, the OS usually distributes different processes on all available processors (CPU) in a way that allows the system to work most efficiently. However, for some reason, you might like to take charge and overrule the kernel's process scheduling to bind your application to a processor/CPU of your choice. This is known as CPU affinity.

In this article, I will try to cover how a normal Linux user can set/retrieve a specific task's affinity from the command line, and then we will go further with the actual implementation of system calls.

So, what is CPU affinity?

On symmetric multi-processing (SMP) systems, the operating system's process scheduler not only decides when a process can run, but also where it should run. CPU (or processor) affinity is the term that describes this property of the scheduler to associate a particular process to a specific processor or CPU.

There are two types of CPU affinity that the Linux scheduler supports: soft affinity (or natural affinity), and hard affinity.

The soft affinity of a process is merely an attempt by the scheduler to run the process on the same processor on which it ran the last time. This way the scheduler tries to improve

